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Palaeomagnetic studies of some Palaeozoic sediments, southwest Sinai, Egypt

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Abstract

Thirty nine sites comprising 301 samples of sandstones and shales of Cambrian, Cambro–Ordovician and Upper Carboniferous age were collected from the Gebel Abu Durba area, southwest Sinai. They mostly show stable magnetisation against AF and/or thermal demagnetisation with a single component of remanence. Both normal and reverse polarities are recorded. Two of the 14 sites of the Cambrian rocks give a palaeopole position at 18.0°N and 341.2°E, which is considered as probably primary. The remaining Cambrian sites give a palaeopole at 29.8°S 56°E, with $A_{95} = 6.7°$. The palaeopole calculated from 11 reliable Cambro–Ordovician sites is at 32.2°S 53.0°E with $A_{95} = 7.9°$. The palaeopole from the 9 Upper Carboniferous sites lies at 25.6°S 64.0°E with $A_{95} = 7.1°$. The obvious similarity of the palaeopoles from most of the Cambrian and Cambro–Ordovician rocks to those of the Upper Carboniferous indicates that all these rocks have a magnetisation of similar age, which, by comparison with other African palaeomagnetic poles, appears to have been acquired during the Upper Carboniferous. Five of the Cambro–Ordovician and Upper Carboniferous sites have normal and reversed polarities and a mean pole of 76.8°N 265.2°E, with $A_{95} = 15.1°$, suggesting a Tertiary remagnetisation associated with the opening of the Gulf of Suez.

1. Introduction

The palaeomagnetic properties of Lower Palaeozoic rocks from Africa, and hence Gondwanaland, remain very poorly defined and controversial. This largely reflects widespread chemical alteration of many of these rocks, which occurred either during the Permian, as reviewed by Creer (1968), or during modern weathering; either of these can have penetrated through hundreds of metres of rocks and are difficult to remove by partial demagnetisation techniques. This is because such chemical remanences affect the total coercivity or blocking temperature spectra. In order to constrain the palaeogeographic history of Gondwanaland further, a study of Palaeozoic rocks exposed on the eastern side of the Gulf of Suez was undertaken in the hope that these rocks may be less affected by these processes as they were only exposed during the predominantly Miocene to Recent faulting associated with the development of the Red Sea and Gulf of Suez.

The Lower Palaeozoic sediments of North Africa are predominantly clastic, with interfingering calcareous sediments, containing few, if any, fossils. In Egypt such occurrences are irregular and widely separated. The lack of diagnostic fos-

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sils makes accurate dating, correlation and formation nomenclature extremely difficult. In Sinai, Palaeozoic sediments are only exposed in the southern central parts (Abdallah and Adindani, 1963; Issawi and Jux, 1982), mainly in the Abu Durba area of southwest Sinai (Fig. 1), where they occur within a long northwest-oriented homocline, which tilts northeastwards with an average dip of 20°. The Palaeozoic sequence comprises 5 m of Cambrian (Omara, 1972), 500 m of Cambro–Ordovician (Said, 1971; Issawi and Jux, 1982; Beleity et al., 1986) and 120 m of Upper Carboniferous rocks (Kora, 1989). Samples were collected from 14 sites, from a series of variously coloured Cambrian calcareous sandstones. These were mainly calcarenites, with fine- to medium-grained, moderate- to well-sorted sandstones. Twelve sites were located in Cambro–Ordovician, white, yellowish to pale brownish, medium- to coarse-grained, well-sorted, quartz arenites. Thirteen sites were also located in moderate- to well-sorted, fine- to medium-grained, quartz arenites, interbedded with fossiliferous shales of Upper Carboniferous age.

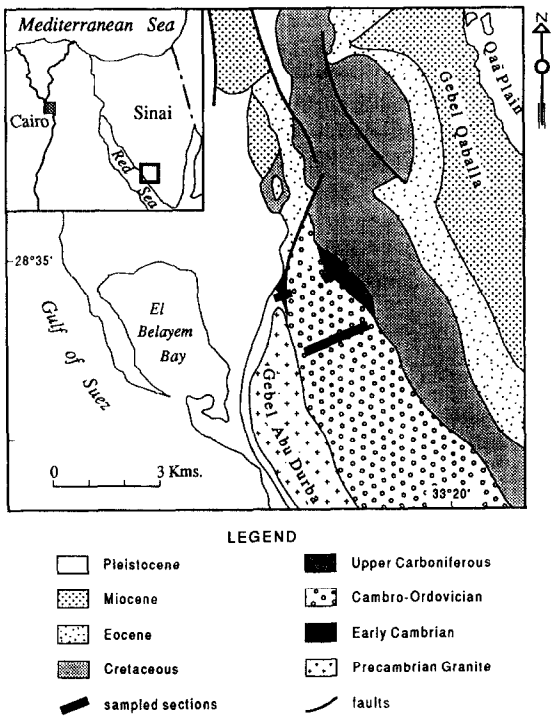


Fig. 1. Simplified geologic map of the Abu Durba area (after Omara, 1972).

Sampling was by field drill and orientations were by sun compass. All measurements were initially carried out with spinner magnetometers (at the Palaeomagnetic laboratory, University of Plymouth), which had a noise level around 0.02 mA/m for standard sized specimens (i.e., 2.5 cm diameter, 2.1 cm high cylinders). Samples for which the signal was close to or below the noise level of these magnetometers were remeasured using the 2 G cryogenic magnetometer at the University of Southampton. Alternating magnetic field demagnetisation was with a Molspin instrument and thermal demagnetisation was with a purpose-built non-magnetic furnace system.

2. Results and analysis

The majority of the Cambrian samples showed low to very low intensities of initial remanence, ranging between 0.11 and 0.55 mA/m and it was necessary to measure samples from 6 sites using the cryogenic magnetometer. The Cambro–Ordovician sandstones showed higher, fairly homogeneous intensities of magnetisation averaging 3.3 mA/m, while the Upper Carboniferous sandstone and shale samples were also somewhat more strongly magnetised than the Cambrian samples, but were more variable, ranging between 0.3 and 29.0 mA/m. The higher intensities were almost entirely associated with samples with a higher shale content. At least one pilot sample from each site was selected for identifying the principal components present and their stability to alternating magnetic fields (AF) applied in incremental steps of 3, 5, 7.5, 10, 15, 20, 25, 30, 40, 50, 60, 70, 80, 90 and 100 mT. All samples showed high stability with no significant directional changes throughout the treatment. The intensity of magnetisation generally fluctuated around its initial values, most of the variation apparently being attributed to demagnetisation 'noise' at such low level of magnetisation. Consequently, the linearities of the individual components were difficult to quantify using the Kirschvink (1980) technique.

As the AF treatment was largely ineffective, it was suspected that the natural remanence was dominated by high-coercivity minerals, probably haematite or goethite. Consequently, a second group of pilot samples was subjected to progressive stepwise thermal demagnetisation at 50°, 100°, 150°, 200°, 300°, 350°, 400°, 500°, 550°, 600°, 650° and 680°C. Bulk susceptibility values were measured at each step so that thermally induced mineralogical changes could be detected. Thermal demagnetisation had distinct effects on the remanences, despite their weakness; confirming that the intensity fluctuations observed during AF treatment were mostly associated with the demagnetisation process rather than with the measurement of their remanences.

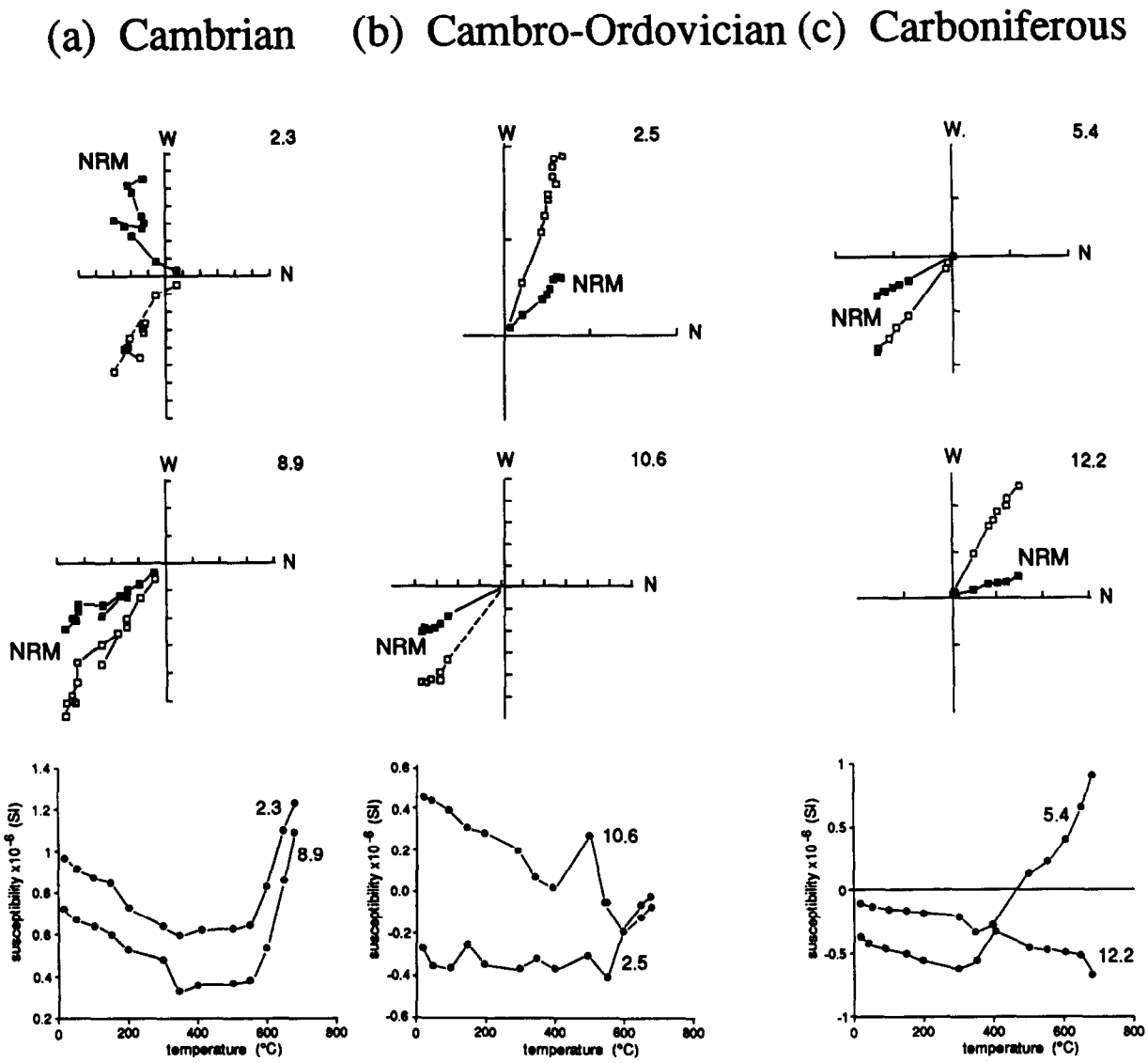


Fig. 2. Examples of vector and low-field susceptibility changes during stepwise thermal demagnetisation. (The directions are corrected for tectonic tilt.) Two examples for each geological group are shown. The vector changes are illustrated on Cartesian plots and the susceptibility changes are shown in graphical form below. On the Cartesian plots, the shaded symbols = the horizontal component; outlined symbols = the vertical component.

All Cambrian samples (example in Fig. 2a), showed a steady decay in remanence until between 500°C and 600°C, above this it tended to increase. The directions of remanence showed no significant changes up to 550°C and were characterised by Stability Indices (SI; Tarling and Symons, 1967) greater than 4; above this temperature began to change randomly. These high-temperature changes seemed to be associated with thermochemical changes induced at high temperatures, rather than the isolation of high-temperature components of natural remanence, as they coincided with changes in the bulk susceptibility and showed no consistency from sample to sample. The Cambro–Ordovician sandstones all behaved similarly (examples in Fig. 2b). Their intensity decayed steadily as heating continued and finally dropped to the magnetometer noise level at 680°C. The directions showed high stability (SI > 15), with no significant directional changes, enabling a well defined, single linear component to be identified, generally with diagonal errors (Kirschvink, 1980) less than 3.0°. Bulk susceptibility values were all very weak and some times negative (diamagnetic), suggesting that the ferromagnetic content was very small. During heating, all sample susceptibilities showed some changes as heating continued, with some showing fluctuating values until temperatures exceeding 500°C (e.g., sample 2.5), while others (e.g., sample 10.6), showed a general decrease, eventually leaving only negative (diamagnetic) susceptibilities above about 400°C (Fig. 2b). These susceptibility changes did not affect the directions of remanent magnetisation and presumably reflect changes in the non-ferromagnetic mineral content, probably clays. Following small initial changes in direction, the remanences of all the Upper Carboniferous sandstones and shales samples showed very high stability ($21.4 \leqslant$ S.I. $\leqslant 42.7$) to thermal demagnetisation (examples in Fig. 2c). Most showed a continuous decay in intensity up to 350°C, above which it tended to increase, and then, above about 400°C, decreased until the magnetisation was completely destroyed by 680°C. Bulk susceptibility also changed at the same temperatures, with some indicating slight mineralogical changes, while others show quite strong increases above 350–400°C. Apart from the first two or three steps, the vectors remained stable and the directions of remanence were unchanged throughout the treatment (Fig. 2c), enabling the isolation of a well defined single characteristic component of magnetisation in the majority of samples.

In order to identify the optimum demagnetisation range within which the characteristic magnetisation was definable, the pilot samples were analyzed statistically, using the Tarling and Symons (1967) Stability Index and the line/plane tests of Kirschvink (1980), and also visually, using both Cartesian and stereographic plots. It was clear that, although the Cambrian samples behaved differently to the two demagnetisation methods, both techniques yielded similar directions. On this basis, alternating fields of 30, 40 and 50 mT were used to treat the remaining Cambrian samples in order to reduce or remove viscous components while avoiding thermochemi-

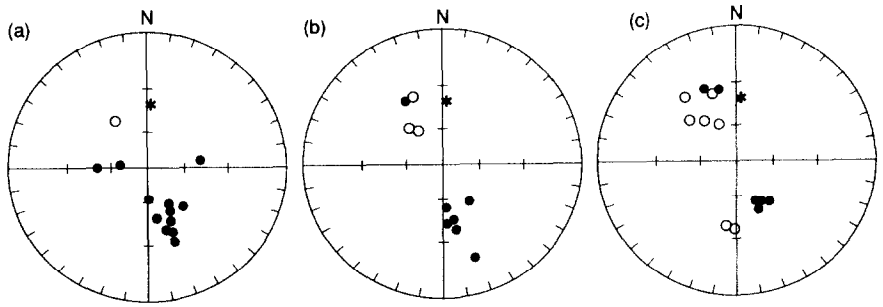


Fig. 3. Equal area stereographic projections of site-mean, cleaned, structurally corrected characteristic NRM directions. (a) Cambrian. (b) Cambro–Ordovician. (c) Upper Carboniferous sites. The directions are listed in Table 1. Star = the direction of the Earth's present magnetic field in the study area.

Table 1
Site mean characteristic NRM and VGPs for the Cambrian, Cambro–Ordovician and Upper Carboniferous rocks

Site	N	Dec. (°)	Inc. (°)	Int. (mA/m)	k	α_{95} (°)	VGP	
							Lat. (°)	Long. (°)
Cambrian rocks								
1	8	272.3	68.1	0.196	16.9	13.9	23.2	350.3
2	8	267.2	50.9	0.422	24.0	11.6	12.3	332.7
Mean	2	269.1	59.5				18.0	341.2
3	8	140.1	48.6	0.431	16.5	14.1	−20.5	69.8
4	8	162.5	39.1	0.237	13.8	15.5	−36.6	53.6
5	7	159.6	36.3	0.205	23.4	12.8	−37.5	57.6
6 [a]	8	81.3	48.2	0.482	13.8	15.5	20.4	100.3
7	6	177.6	62.8	0.221	28.9	12.7	−17.2	35.1
8	7	150.6	54.2	0.514	30.9	11.0	−20.9	58.9
9	7	153.9	50.5	0.273	20.4	12.6	−25.3	57.8
10	8	157.7	43.4	0.244	46.9	8.2	−32.1	57.1
11	8	158.5	36.2	0.346	45.2	8.3	−37.1	58.8
12	8	160.4	31.3	0.317	43.5	8.5	−40.8	58.3
13	8	169.7	48.7	0.491	22.3	12.0	−31.0	43.7
14	8	326.7	−46.8	0.212	41.1	8.8	25.1	245.6
Mean	11	157.6	45.6				−29.8	56.0
Cambro–Ordovician rocks								
1	7	169.4	43.2	3.470	50.4	8.6	−35.4	45.0
2	9	336.8	−34.8	0.873	39.2	8.3	37.2	241.2
3	8	173.2	41.9	1.110	244.3	3.6	−36.9	41.0
4 [b]	7	330.1	34.4	0.239	27.6	11.7	61.1	290.5
5	7	167.6	37.1	2.196	110.6	5.8	−39.3	48.3
6	9	161.5	18.2	4.876	123.4	4.7	−48.1	61.2
7	8	169.6	43.9	1.099	54.6	7.6	−34.9	44.7
8	8	174.6	52.9	1.917	25.6	11.2	−27.8	38.4
9	8	324.8	−57.7	2.311	98.3	5.6	15.5	241.2
10	7	317.4	−51.7	0.383	21.3	13.4	16.9	250.0
11	7	170.2	42.0	0.499	67.5	7.4	−36.4	44.4
12	7	145.8	52.2	4.756	262.2	3.7	−20.6	63.6
Mean	11	161.9	43.9				−32.2	53.0
Upper Carboniferous rocks								
1	7	341.3	−35.9	1.620	23.7	12.7	38.3	235.8
2	8	322.0	−49.5	0.730	61.2	7.1	20.9	247.9
3 [b]	7	347.1	34.9	5.117	148.6	5.0	74.9	267.4
4	7	157.4	47.4	17.507	66.0	7.5	−29.0	56.0
5	7	143.6	48.3	28.589	153.6	4.9	−22.5	67.3
6 [b]	8	189.6	−38.0	13.309	186.5	4.1	−78.7	340.5
7 [b]	8	336.7	31.0	14.897	291.6	3.3	65.5	279.4
8	7	321.7	−29.3	1.152	288.5	3.6	32.3	258.2
9	7	311.8	−42.3	3.079	70.9	7.2	19.5	259.4
10	7	155.3	53.2	6.203	994.1	1.9	−23.5	55.5
11	8	148.7	50.7	6.939	423.3	2.7	−23.3	62.0
12	8	335.5	−58.2	0.910	221.9	3.7	18.8	233.2
13 [b]	8	181.8	−36.7	0.224	26.6	10.9	−81.8	21.2
Mean	9	148.3	46.5				−25.6	64.0
Mean [b]	5	348.7	35.9				76.8	265.2

N = number of samples in each site (or sites for mean calculation).
Dec. and Inc. = structurally corrected magnetic declination and inclination.
Int. = intensity of magnetisation.
k = precision parameter (Fisher, 1953).
α_{95} = semi-angle cone of confidence (Fisher, 1953).
[a] Site excluded from overall mean calculations.
[b] Sites combined for mean value at bottom of the table.

cal changes. For the remaining Cambro–Ordovician samples, three successive thermal steps of 300°, 350° and 400°C were chosen for removing soft and viscous magnetisations and leaving a measurable stable characteristic magnetisation. A slightly different temperature sequence, 200°, 300° and 350°C, was used for the remaining Upper Carboniferous samples. Linear and stable components isolated in these different ranges were then combined with those already evaluated for the pilot samples in order to determine the characteristic components of remanence.

The bulk demagnetisation generally reduced the within-site scatter in most sites and also improved the between-site grouping. In some sites, individual sample directions were deviant (i.e., more than 40° from the mean) and these were excluded in determining the mean site directions and corresponding palaeomagnetic pole positions (Table 1). After correction to bedding, the majority of the sites in each of the three groups displayed declinations pointing either to the SSE with moderately steep positive inclinations or to the NW, with relatively steep negative inclinations, well away from the Earth's present magnetic field direction (Fig. 3). The individual age group characteristics were:

(1) The site mean directions of 11 of the Cambrian reliable sites showed a clear grouping, with one antiparallel magnetisation (site 14). With this antiparallel site direction reversed, these sites have an overall mean direction, giving unit weight to each site and after correction for tilting, of Dec. = 157.6°, Inc. = 45.6° ($k = 52$; $\alpha_{95} = 6.4°$), with a corresponding palaeomagnetic pole at 29.8°S 56.0°E ($A_{95} = 6.7°$). Two other Cambrian site mean directions (1 and 2) were similar to each other, although 17.4° apart, and had a mean direction of Dec. = 269.1° and Inc. = 59.5°, corresponding to a palaeomagnetic pole of 18.0°N 341.2°E. Only one site (6) did not belong to either group.

(2) Eleven of the Cambro–Ordovician site directions were similar after reversing 3 antiparallel sites (2, 9 and 10) and these then had an overall mean direction of Dec. = 161.9°, Inc. = 43.9° ($k = 34$; $\alpha_{95} = 8.0°$), with a corresponding palaeomagnetic pole at Lat. = 32.2°S, Long. = 53.0°E ($A_{95} = 7.9°$). One site (4) was apparently reversed in declination, but not in inclination (discussed below).

(3) Four of the Upper Carboniferous sites (3, 6, 7 and 13) were similar to each other, but with site mean directions in sites 3 and 7 similar to that of Cambro–Ordovician site 4, and antiparallel to those of sites 6 and 13. The mean direction of these 5 sites is Dec. = 348.7°, Inc. = 35.9° ($k = 35$; $\alpha_{95} = 13°$) with a corresponding palaeomagnetic pole at 76.8°N, 265.2°E ($A_{95} = 15°$). This direction, which is similar to the Earth's present magnetic field (2.6°, +41.4°) could suggest a Recent magnetisation, but the presence of two reversed sites would suggest a Tertiary age, probably remagnetisation associated with the time of faulting of the Gulf of Suez. The remaining Upper Carboniferous sites had a mean direction of Dec. = 148.3°, Inc. = 46.5° ($k = 53$; $\alpha_{95} = 7.2°$) and a pole position of 25.6°S 64.0°E ($A_{95} = 7.1°$), after reversing 5 site directions.

A reversal test (McFadden and McElhinny, 1990) was applied to the site mean directions from the Cambro–Ordovician and the Upper Carboniferous rocks in order to test whether their normal and reversed directions (Table 1) could have been drawn from the same population. The angle between the mean of the two polarity sets of the Cambro–Ordovician sites, after changing polarity of one group, was found to be 15°, while the critical angle was 19.4°. The angle between the two sets of polarities for the Upper Carboniferous sites was 7.4°, while the critical angle was 13.6°. Under the null hypothesis, both the Cambro–Ordovician and Upper Carboniferous site directions are therefore drawn from populations with means that are 180° apart and their directions could be resolved to better than 20° but not better than 10°. The classification in both cases is therefore positive ('C' category; McFadden and McElhinny, 1990).

3. Interpretation and conclusions

There is a clear consistency between most of the site mean directions, irrespective of age, sug-

gesting that these sites were magnetised at a similar time and, hence, that most of the Cambrian and Cambro–Ordovician rocks, at least, have been completely and uniformly remagnetised—with two notable exceptions (discussed later). The fact that these mean directions differ very clearly from the present and Late Tertiary geomagnetic field, and that reversals are observed, demonstrates that this is neither a viscous nor a chemical remanence acquired in the last few million years. The 22 Cambrian and Cambro–Ordovician site means that are similar to each other have a mean direction of 159.8°, 44.8° ($k = 42$; $\alpha_{95} = 4.8°$) and a mean palaeomagnetic pole at 31.0°S 54.5°E ($A_{95} = 5.4°$). This is only 8.2° from that of the mean Upper Carboniferous mean site direction of 148.3°, 46.5° ($k = 53$; $\alpha_{95} = 7.2°$) and, thus, the two means cannot be considered statistically different at a 95% confidence level. The problem is then whether the Upper Carboniferous rocks carry a primary magnetisation or were also remagnetised, with the older rocks, at some later time. Unfortunately, neither the Upper Carboniferous nor Triassic African poles are sufficiently well defined in either magnetic properties or age of magnetisation to enable precise assessment by comparison. However, the general similarity with other African Upper Carboniferous pole positions (Table 2, Fig. 4b) suggests that an Upper Carboniferous age is most likely.

The two remaining Cambrian poles must clearly be treated cautiously, as they are not identical to each other, however, they are both close to some African poles of Ordovician age (Fig. 4a). This could suggest that they have suffered an Ordovician remagnetisation. However, there is no evidence for such a remagnetisation and they are also close to the very well defined Ntonya Complex pole (Briden, 1968) that has recently been re-dated as being of mid-Cambrian age (Briden et al., 1993), suggesting that these two sites have retained their original Cambrian magnetisation and thus confirming the general location for the African Cambrian pole position off northwestern Africa. However, such a position is inconsistent with the palaeoclimatic evidence for very low palaeolatitudes at this time in Morocco: possibly reflecting an allochthonous nature for this part of North Africa at this time.

Table 2
Cambrian (ϵ), Ordovician (O) and Carboniferous (C) Palaeomagnetic South Poles for Africa

Rock unit	Region	Age	Lat.	Long.	A_{95}	References
1	Hassi-Bechar RB, Algeria	Cu	−26.8	56.6	3.7	Daly and Irving, 1983
2	Ain-Ech Chebbi RB, Algeria	Cu	−23.0	52.0	6.0	Daly and Irving, 1983
3	Dywka glacial varves, Cent. Afr.	Cu	−26.5	26.5	10.5	McElhinny and Opdyke, 1968
4	Abu Durba sediments, Egypt	Cu	−25.6	64.0	7.1	This study
5	Carboniferous L.S, Morocco	Cu	−7.4	54.5	11.0	Salmon et al., 1988
6	Carboniferous rocks, Morocco	Cl	−4.8	55.5	6.1	Martin et al., 1978
7	Sabaloka Complex, Sudan	Cl	−42.0	55.7	7.0	Soffel et al., 1990
8	Salala Complex, Sudan	Om	39.6	329.5	9.3	Bachtadse and Briden, 1989
9	Graafwater Fm, South Africa	Ol	28.0	14.0	8.8	Bachtadse et al., 1987
10	Ntonya Complex, Malawi	ϵm	28.0	345.0	5.0	Briden, 1968, Briden et al., 1993
11	Nabati Complex, Sudan	ϵl	67.9	314.1	13.6	Saradeth et al., 1989
12	Abu Durba sediments, Egypt	ϵl	18.0	341.2	44.9	This study

See also Fig. 4. The published pole positions have been selected on the following criteria: (1) the number of sites used is not less than 5; (2) demagnetisation treatment was applied (AF, thermal or chemical); (3) the magnetic age must not exceed a geologic period; (4) the confidence limit, α_{95}, is $< 15°$; (5) remagnetisation was not reported by the authors; (6) no local tectonic rotation was proposed by the authors.
Poles from the islands around Africa (e.g., Sicily, Canary Islands and in the Gulf of Guinea) are also excluded as their tectonic relationship to Africa is not established.

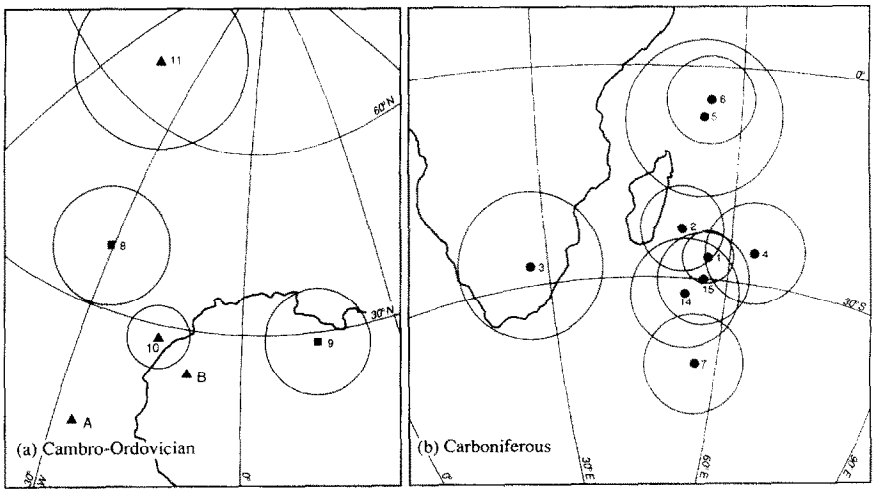


Fig. 4. African Palaeozoic pole positions. Previously published and the new Sinai south palaeomagnetic pole positions (Table 2) show (a) the Cambrian and Ordovician and (b) anomalous poles grouped around the Carboniferous pole positions for Africa. The ages of the rocks involved are Cambrian (triangles), Ordovician (squares) and Carboniferous (dots). The remagnetised poles for the Abu Durba Late Palaeozoic sediments, attributed to the Lower Carboniferous, are poles 14 and 15. Both the Cambrian site poles defined in this study, A and B, are shown rather than their average as their circle of confidence, based on only 2 values, is large (A_{95} = 44.9°). The projection is stereographic.

The observations from this area thus contain apparently primary magnetisations in 2 of the Cambrian sites and 9 of the Upper Carboniferous sites. There is also evidence for Late Palaeozoic remagnetisations of 22 of the Cambrian and Cambro–Ordovician sites and 5 of the Cambro–Ordovician and Upper Carboniferous sites show apparently Tertiary remagnetisations. It is interesting to note that these remagnetisations are quite discrete; that is, no samples show two components of magnetisation, suggesting that the remagnetisations are most likely of chemical origin and also clearly spatially defined. Such exposures therefore afford the opportunity for retrieving some Palaeozoic remanences and also for studying the nature and zonation of some of the events associated with their tectonic evolution.

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